🧭 Cyber Dash Launcher – Updated Usage & Deployment Guide

(Fully Controlled by launch.env — No Batch or PowerShell Scripts Needed)

⚙️ 1. Overview

This launcher (open-dash.exe) is a lightweight Windows utility built in C to open a **dashboard web interface** (such as your VLAN or cyber range portal).
It reads configuration variables from launch.env — not hardcoded IPs — making it easy to switch between VLANs or networks.

Key Points:

- No PowerShell, batch, or admin privileges required.
- Single .exe file controlled by editable text file (launch.env).
- Portable and network-aware — ideal for classroom, demo, or cyber lab use.

🧩 2. Project Directory Structure

```
DashboardLauncher/
├── assets/
│   └── app.ico        # Application icon used during build
├── build.sh           # Bash build script (used via WSL or Linux)
├── icon.rc            # Resource compiler script (references app.ico)
├── open_dash.c        # Source code (C)
├── launch.env         # Environment configuration (edit to change settings)
└── open-dash.exe      # Final executable (output after build)
```

🧰 3. Files Explained

File	Purpose
open_dash.c	Main C program that reads configuration and launches dashboard URL.

File	Purpose
launch.env	Configuration file that defines IPs, ports, and behavior (ping delay, sleep, etc).
build.sh	Script that compiles open_dash.c and resources into a Windows .exe (run inside WSL).
icon.rc	Defines the icon resource used by Windows when displaying the EXE.
assets/app.ico	The program’s icon, generated from your original .png file.
open-dash.exe	The compiled launcher program — ready for distribution.

🧾 4. Example launch.env

DASH_URL=http://172.20.10.30:8000/

DASH_PING_HOST=172.20.10.1

DASH_PING_COUNT=2

DASH_SLEEP_MS=1500

DASH_DO_PING=1

🧠 **How it works:**

- Each key-value pair defines an environment variable.
- The launcher reads these at startup.
- If a value is missing, the program falls back to safe defaults.

🚀 5. How to Use

1. **Place both files** in the same directory:
2. open-dash.exe
3. launch.env
4. **Edit launch.env** to point to the desired web server:

5. DASH_URL=http://192.168.5.30:8000/

💡 No recompilation needed. Just edit and save.

6. **Double-click open-dash.exe.**
   - The launcher will read launch.env.
   - Ping the configured host (optional).
   - Sleep for the defined delay.
   - Open your dashboard in the default web browser.

🔁 6. To Edit / Update / Rebuild

To edit configuration:

Simply modify launch.env and save.
That's it — the executable automatically respects the new IP and parameters.

To recompile (if you change the code or icon):

Run these commands inside WSL (Ubuntu):

cd /mnt/c/Users/<YourName>/Desktop/DashboardLauncher

chmod +x build.sh

./build.sh

🧱 7. Deployment in a Classroom or Lab

Scenario	Action
Switch VLANs (e.g., Jumpbox, 10.10.10.0/24 to 172.20.10.0/24)	Edit DASH_URL and DASH_PING_HOST in launch.env.
Deploy to another PC	Copy only open-dash.exe and launch.env. No setup required.
Portable Drive / USB	Store both files in one folder — plug and launch from any Windows 10/11 system.

🧠 8. How It Works Internally

When you launch open-dash.exe, it does the following:

1. Reads launch.env line by line.
2. Sets those environment variables for the current process using _putenv_s().
3. Pings your gateway (optional awareness feature).
4. Waits (Sleep() delay).
5. Launches your default browser and opens the DASH_URL.

No registry entries. No background processes. No PowerShell.

🧰 9. Troubleshooting

Problem	Fix
Browser doesn't open	Check that DASH_URL is correctly formatted with http://
Ping errors	Verify DASH_PING_HOST is reachable or set DASH_DO_PING=0
Nothing happens	Ensure launch.env is in the same directory as open-dash.exe
Antivirus alert	Code is unsigned — optionally sign with your internal certificate authority before deployment.

🪪 10. Security & Ethics Note

This project is a **security awareness and automation tool** — not an exploit.
It demonstrates:

- Controlled automation using environment-based configuration.
- Practical awareness of network separation (VLANs, jumpboxes).
- Reproducible builds for academic and ethical training.

Would you like me to generate this section as a .md file (for upload directly into your GitHub repo DashboardLauncher/README.md)?